CS-302 Distributed Systems - Week 6 Lecture Notes

Topic: Raft Consensus Algorithm

Raft elects a single leader per term. Candidates request votes from peers and win with a majority quorum. Heartbeats flow from the leader to followers every 100-300ms. If a follower misses heartbeats it starts a new election with an incremented term number.

Log replication: the leader appends entries and replicates them to followers. An entry is committed once a majority acknowledge it. Commit index advances monotonically.

CS-302 Distributed Systems - Week 6 (continued)

Topic: Vector Clocks and Causality

Vector clocks capture causality across replicas. Each process maintains a vector of counters, incremented on local events and merged on message receipt.

Exam hint: expect a question comparing Raft leader election with the failure detector approach in Paxos. Review the midterm scope page on the course site.